



Oxford Cambridge and RSA

Friday 6 June 2025 – Afternoon

AS Level Further Mathematics A

Y535/01 Additional Pure Mathematics

Time allowed: 1 hour 15 minutes



You must have:

- the Printed Answer Booklet
- the Formulae Booklet for AS Level Further Mathematics A
- a scientific or graphical calculator

QP

INSTRUCTIONS

- Use black ink. You can use an HB pencil, but only for graphs and diagrams.
- Write your answer to each question in the space provided in the **Printed Answer Booklet**. If you need extra space use the lined page at the end of the Printed Answer Booklet. The question numbers must be clearly shown.
- Fill in the boxes on the front of the Printed Answer Booklet.
- Answer **all** the questions.
- Where appropriate, your answer should be supported with working. Marks might be given for using a correct method, even if your answer is wrong.
- Give non-exact numerical answers correct to **3** significant figures unless a different degree of accuracy is specified in the question.
- The acceleration due to gravity is denoted by $g \text{ ms}^{-2}$. When a numerical value is needed use $g = 9.8$ unless a different value is specified in the question.
- Do **not** send this Question Paper for marking. Keep it in the centre or recycle it.

INFORMATION

- The total mark for this paper is **60**.
- The marks for each question are shown in brackets [].
- This document has **4** pages.

ADVICE

- Read each question carefully before you start your answer.

- 1 Use standard divisibility tests for small numbers to show that 2 573 208 is divisible by 792. [4]
- 2 The vectors $\mathbf{p}$, $\mathbf{q}$ and $\mathbf{r}$ are such that $\mathbf{p} = \begin{pmatrix} 1 \\ 3 \\ 7 \end{pmatrix}$, $\mathbf{q} = \begin{pmatrix} 2 \\ -4 \\ 9 \end{pmatrix}$ and $\mathbf{r} = \begin{pmatrix} 11 \\ 1 \\ -2 \end{pmatrix}$.
- (a) (i) Determine the value of the integer k for which $\mathbf{p} \times \mathbf{q} = k\mathbf{r}$. [2]
- (ii) Use the result of part (a)(i) to explain **geometrically** why $(\mathbf{p} \times \mathbf{q}) \times \mathbf{r} = \mathbf{0}$. [1]
- Relative to the origin O , the points P and Q have position vectors $\mathbf{p}$ and $\mathbf{q}$ respectively.
- (b) Show that the area of triangle OPQ can be written in the form $\frac{1}{2}n\sqrt{n-1}$ where n is a positive integer to be determined. [3]
- 3 The surface S has equation $z = 2x^2y - 6y^3 + 3x + 4$ for all real values of x and y .
- (a) (i) State the equation of the section of S cut by the plane $y = 1$. [1]
- (ii) Sketch the section of S cut by the plane $y = 1$. Give the coordinates of the points of intersection with the axes. [2]
- (b) Determine the coordinates of all stationary points of S . [6]
- (c) The contour C of S is given by $z = 37$.
- Find the coordinates of the unique point on C where $\frac{\partial^2 z}{\partial y^2} + \frac{\partial^2 z}{\partial y \partial x} = 0$. [4]
- 4 The binary operation $*$ is defined on the set $A = \{1, 3, 5, 7, 9\}$ by $x * y = x + y + 3 \pmod{10}$ for all $x, y \in A$.
- (a) (i) Show that $*$ is associative on the elements of A . [2]
- (ii) Complete the Cayley table for $(A, *)$ given in the Printed Answer Booklet. [2]
- (iii) Hence show that $(A, *)$ forms a group, G . [4]
- (b) (i) State the order of each non-identity element of G . [1]
- (ii) List all subgroups of G , and explain why there are no others. [2]
- (iii) Explain whether G is cyclic. [1]
- (iv) Explain whether G is abelian. [1]

- 5 Let $f(n) = 3^n + 4^n$ for all positive integers n .

Prove by induction that $f(n)$ is a multiple of 7 for all **odd** integers $n \geq 1$. [5]

- 6 An investment company offers customers a three-year investment scheme.

The scheme is modelled by the recurrence system

$$I_0 = \alpha \text{ and } I_{n+1} = 0.000625 I_n^2 - 0.625 I_n \text{ for } n \geq 0.$$

I_n is the value in pounds of the scheme n years after its start, and α is the **integer** value of the initial sum invested.

- (a) Determine the minimum initial sum invested that guarantees that the value of the scheme increases every year. [3]

- (b) The investment company decides to make the following change in the scheme.

At the end of each year, the value of I_n is always rounded **down** to the nearest pound.

- (i) Write down a modified recurrence formula that takes this change into account. [1]

- (ii) A customer invests £2700 in the three-year investment scheme.

Work out how much less the customer's investment would be worth after three years under the new scheme compared to the old scheme. [6]

- 7 Consider the two arithmetic sequences $A_n = \{2n+7: n \in \mathbb{N}\}$ and $B_n = \{3n+1: n \in \mathbb{N}\}$. Let $h = \text{hcf}(A_n, B_n)$ for each chosen value of n .

- (a) (i) Find a value of n for which $h = 1$. [1]

- (ii) Find a value of n for which $h = 19$. [2]

- (b) Determine all the values of n for which $h = 19$. Give your answer as an expression in terms of an integer k . [2]

- (c) By considering $3A_n - 2B_n$, show that A_n and B_n are co-prime for all values of n that are not given by the expression found in part (b). [4]

END OF QUESTION PAPER

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